ZOO103 CURRENT PAPERS

SPRING 2022

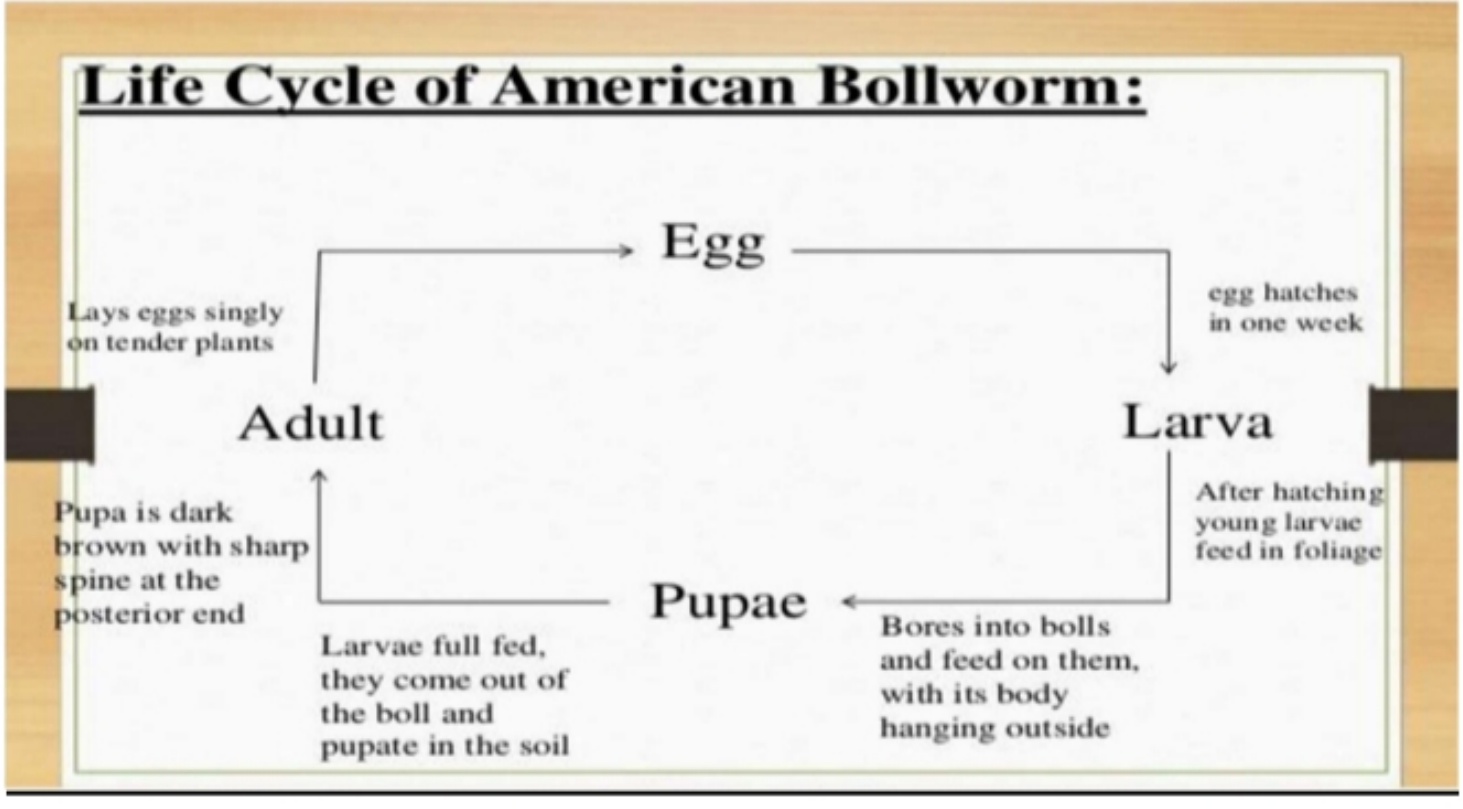
FINAL TERM

EFFORT BY HUMA

Q:-General characteristics of Phylum Coelenterata 5

1. Coelenterata is also known as Cnidaria.
2. This name of cnidaria is due to the single cavity Coelenteron.
3. Radial Symmetry
4. Diploblastic
5. No separate body cavity.
6. They have an incomplete digestive systems. It has a single aperture mouth but doesn't have an anus.
7. They respire through the body surface by diffusion.
8. They excrete through body surface by diffusion.
9. Segmentations are absent in them.
10. They are mostly marine, aquatic animals and free living animals.

Life Cycle of American Bollworm:



Classification of green lacewings

Green lacewings are an often under-appreciated group of beneficial insects.
The adult green lacewing is about 3/4 inch long, light green and has a delicate appearance with lacy wings. • One unusual characteristic are its eyes, they look like two golden hemispheres. • They are weak fliers and are commonly found near aphid colonies.

Biogeography· Evolutionary biology

sympatric speciation: • A speciation in which a species is evolved from a single common ancestral species while inhabiting the same geographical range.• Share the same habitat but become reproductively isolated from each other.• More common in plants

Arthropods legs parts: Coxa• Traochant• Femur • Patella• Tibia• Metatarsu• Tarsus• Claws

Composition of silkworm.

silkworm consists of two main proteins, sericin and fibroin, fibroin being the structural center of the silk, and serecin being the sticky material surrounding it.

Chemical Composition of Silk Fiber: · Fibroin: 75% · Ash of Silk Fibroin: 0.5% · Sericin: 22.5% · Fat and Wax: 1.5% · Mineral Salt: 0.5%

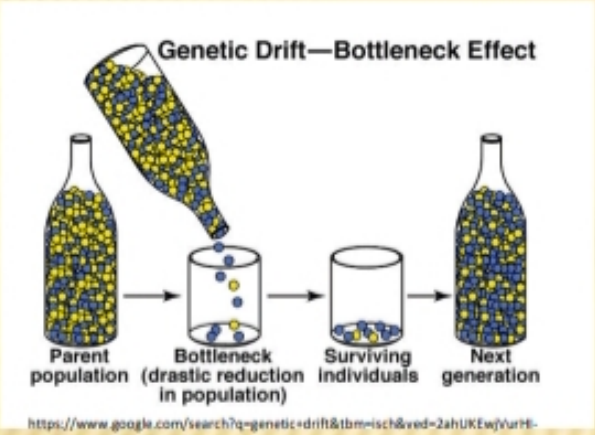
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Lac Insect

Kingdom	Animalia
Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Kerriidae
Genus	Kerria
Species	*Kerria lacca*

- Lac insect live as parasite on sap of certain tress and shrubs and shows sexual dimorphism
- *Babul (Acacia arabica)*
- *Khair (Acacia catcchu)*
- Its life is completed in 6 months
- Two crops can be obtained

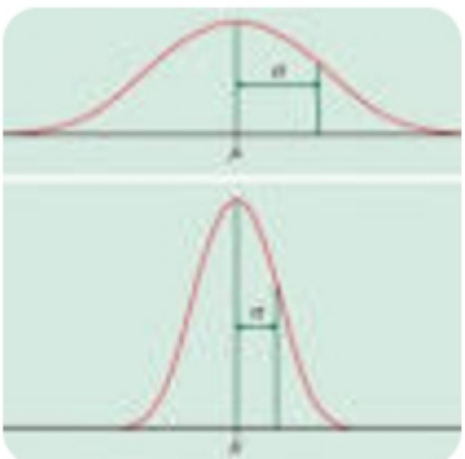
Genetic Drift and its impact on Allele Frequency



https://www.google.com/search?q=genetic+drift&tbm=isch&ved=2ahUKEwjVurHi-evtAhVPw4UKHYbCBvoQ2-cCegQIABAA&oq=genetic+&gs_lcp=CgNpbWcQARgAMgQIABBDMgQIABBDMgQIABBDMgIIADICCAAyAggAMgIIADICCAAyAggAMgIIADoFCAADsQM6BAgjECc6BwgAELEDEENQ3KsjWNie6I2D6zCNoAHAAwACAAAfUDIAGCESIBCTIrNi4wLjEuMZgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=BVXnXSWkA8-GIwSGhZvQDw&bih=657&biw=1360#imgrc=bg-AkcfJaUx8GM

- Change in allege frequency which occur by chance.
- Loss in alleles
- Loss in genetic diversity
- Lost genes for ever
- Bottle neck effect
- Unique genes
- Examples

What a standard deviation means?



A standard deviation (or σ) is **a measure of how dispersed the data is in relation to the mean**. Low standard deviation means data are clustered around the mean, and high standard deviation indicates data are more spread out.

Non-Parametric Tests

- Non-parametric tests for skewed data.
- Wilcoxon Signed Rank Test
- Mann Whitney U Test
- Kruskal-Wallis test
- Principal Component Test
- Friedman Tests

Three types of sponges

The approximately 5,000 living sponge species are classified in the phylum Porifera, which is composed of three distinct groups, the Hexactinellida (glass sponges), the Demospongia, and the Calcariea (calcareous sponges).

[redacted]

Parts of thesis & Research paper

Parts of references

- Name of authors
- Year of publication
- Title/topic of article
- Name of Journal
- Volume number
- Issue number
- Page number

Systematic Sampling

- In this sampling method first sample from the population is taken randomly but after that samples are selected systematically at a regular interval.

Examples:

- Shopping centre
- Selection of plants
- Selection of soil sample

- **Quota sampling:** In this type of sampling we divide the population intro different groups or strata based on some specific criterion but selection of individual from each group is not random.

ANOVA

- Analysis of variance (ANOVA) is a technique used to compare means of more than two samples.
- This technique can be used only for numerical data as it is a parametric test."

ANOVA

Assumptions

- Number of samples must be more than two.
- Data should be normally distributed.
- Samples must be independent.
- Homogeneity of variances.

ANOVA

Examples

- A group of psychiatric patients are trying three different therapies: counseling, medication and biofeedback. You want to see if one therapy is better than the others.
- Students from different colleges take the same exam. You want to see if one college outperforms the other.

END

SOCIAL VARIATIONS

Social variation in ants

- Ants live in complex social colonies.
- **Queen:** founder and leader of the colony.
- Primary goal is to populate colony .
- Lay thousands of eggs
- Queen chambers are deep within the colony for protection.
- Long life in some species upto 30 years.

SOCIAL VARIATIONS

Social variation in ants

- **Drones:** Only function is to fertilize female
- Die after fertilization
- Rarely encountered outside the colony
- **Workers:** Foraging and protection
- Sometime: queen lay eggs, winged male and females-**Alates or reproductive.**

SOCIAL VARIATIONS

Social variation in ants

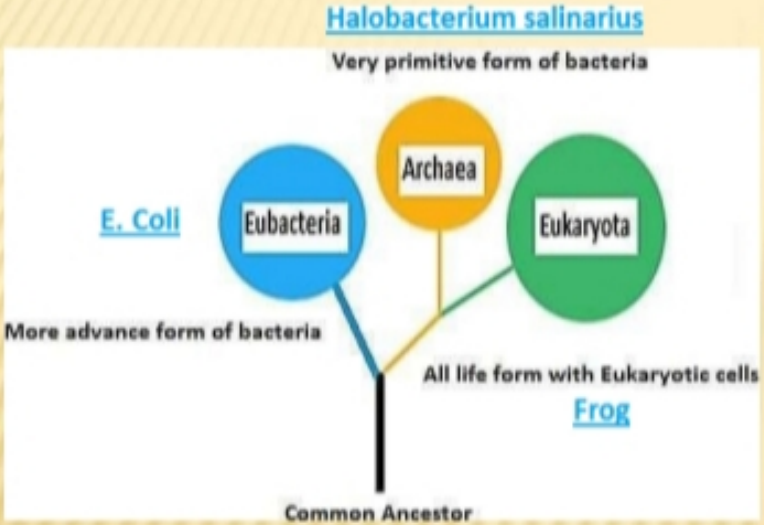
- After maturation leave the colony in swarms.
- Establish new colony

[redacted]

Conditions for t test

The conditions required to conduct a t-test include the measured values in ratio scale or interval scale, simple random extraction, homogeneity of variance, appropriate sample size, and normal distribution of data.

THREE DOMAIN SYSTEM



- A domain is a broadest taxonomic category.
- All organism belong to one of the three domains, depending upon their characteristics.
- A single domain can contain one or more kingdoms.
- Introduced by Charl Woese et al. In 1990.

aphid

/ˈeɪfɪd/

noun

plural noun: **aphids**

a small bug which feeds by sucking sap from plants; a blackfly or greenfly. Aphids reproduce rapidly, sometimes producing live young without mating, and large numbers can cause extensive damage to plants.





Note on sericulture in Pakistan

Despite the growing demand for natural silk, this industry could not flourish in Pakistan.

• This industry is facing numerous challenges such as lack of trained personnel, high mortality rate in local silk breed, and poor economic conditions of the farmers. • Unfortunately, this industry has not been on the list of priorities of us Government.

• Provide livelihood for rural poor families but silk proteins also have wide range of applications in textile, food, cosmetics and pharmaceutical industry as well as in [illegible] medicines.

• Currently, Pakistan is importing about 70% raw silk to meet the demand of [illegible]

[redacted]

characteristic of spiders 5

1. These characteristics include bodies divided into two tagmata (sections or segments),
2. eight jointed legs,
3. no wings or antennae,
4. the presence of chelicerae and pedipalps,
5. simple eyes, exoskeleton, which is periodically shed.

Characteristics of the Phylum Arthropoda

- The segmented bodies are arranged into regions, called tagmata (e.g., head, thorax, abdomen).
- The paired appendages (e.g., legs, antennae) are jointed.
- They posses a chitinous exoskeletion that must be shed during growth.
- They have bilateral symmetry.
- The nervous system is dorsal (belly) and the circulatory system is open and ventral (back).

CORRELATED CHARACTERS

- Characters which are functionally correlated have very low weight
- They have low phylogenetic content
- Do not deserve to be treated as separate character
- Analogues organs birds wings, bats and insects (analogus)/convergent evolution
- Redundant

Evolution

- Evolution is a process by which plants and animals that exist today have been evolved from plants and animal of past.
- Before the rise of genetics, the study of evolution was carried out entirely by taxonomists.
- Virtually all major evolutionary problems were first pointed out and often solved by systematics.

Random Sampling

- If every individual of target population has an equal chance of selection then it will be called random sampling. Simple random sampling is the easiest form of probability sampling.
- No biasness and expert opinion is required in this method.

END

Random Sampling

- All the researcher needs to do is assure that all the members of the population are included in the list and then randomly select the desired number of subjects.

Examples:

- Selection of plants from rice fields
- Selection of students from class
- When carrying out a survey, it would be impractical to study a whole population.

- **Poison Distribution**
- The data distribution in which total numbers are not predetermined and some time there number can reach to infinite. These events occur in nature rarely. Outcome of such events is complete discrete value not in decimal points. For example number of deaths in Lahore per hour. Number of deaths can be infinite if some unusual event occurs.



Swarming

- These larvae begin to wander in search of suitable centre to fix them.
- This mass movement of larvae from female cell to the new off-shoots of host plant, is termed as "swarming".

Cluster Sampling

- It is a sampling technique where the entire population is divided into groups, or clusters and a random sample of these clusters are selected.
- Members of one cluster have similarity with each other and also with other clusters; minor differences can be there between different clusters.
- Individual selection from each cluster is random.

Cluster Sampling

- Cluster of plants for study
- Cluster of animals

predator

/ˈprɛdətə/

See definitions in:

All | Zoology | Commerce

noun

1. an animal that naturally preys on others.
   "wolves are major predators of small mammals"

- Monographs try to review all the facts about a taxonomic group.
- A **Monograph** is a complete global account of a **taxon** of any rank – family, genus or species at a given time.
- Book format

What is water vascular system explain?

Definition of water-vascular system

: **a system of canals in echinoderms containing a circulating watery fluid that is used for the movement of the tentacles and tube feet**.

https://www.merriam-webster.com › ...

Water-vascular system Definition & Meaning - Merriam-Webster

More results

Do all echinoderms use the water vascular system?

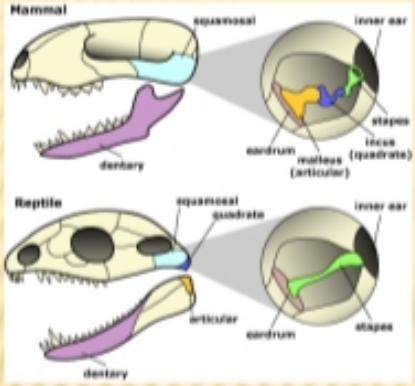
It is a characteristic feature of all echinoderms. It includes a system of canals within the body in which the seawater circulates. The water vascular system includes a ring canal in the central part of the body.

Biological Species Concept

- **"A species is a group of interbreeding natural population that is reproductively isolated from other such species"**
- A species has three properties:
- Reproductive community
- Ecological Unit
- Genetic Unit

karyology (plural karyologies) (biology, medicine) **The study of the nuclei of cells, especially with regard to the chromosomes which they contain**.
(biology, medicine) The characteristics of a particular cellular nucleus or group of nuclei, or of the chromosomes therein.

ANALOGY



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- The structure which are different in different species but perform similar function.
- Have similar appearance
- Not evolved together
- Do not share common ancestor
- Linnaeus started classification on the basis of morphological similarities.
- Wrong classification
- All similar looking organism do not have same origin

Mimicry





Camouflage ➡
- To change color of body to shade into scenery

Mimicry ➡
- To take the form of other organism, such as plant or insects

https://www.slideshare.net/m5g7j8/mimicry-70192874

- Mostly in mimictic relationship advantage is one side.

Types of Mimicry

- Batesian mimicry
- Mullerian mimicry
- Wasmannian mimicry
- Peckhmian mimicry
- Self-mimicry

Which is called flora and fauna with examples?

Flora: **flora means the plants naturally occurring in a particular area**. Some examples of flora include- grasslands, forests, flowering and non-flowering plants and trees. Fauna: fauna means the animals naturally living in that area. Few examples of fauna include- birds, animals, fish, insects, etc.

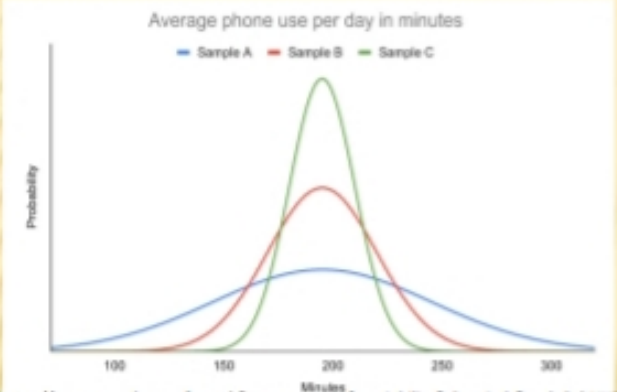
Explanation: The gizzard is a specialised structure found in the stomach of birds that have thick muscular walls, and is **useful for grinding food, stones and grit**. Crop, on the other hand, is a thin-walled organ used for the storage of food prior to digestion.

Measure of Central Tendency and Measure of Variability

Measure of central tendency

- It provide information about average value of a data set.
- Mean, Median and Mode are commonly used measures of central tendency.
- Data set has only one and only one Mean.
- Value of mean is influenced by all observations in the data set.

Measure of Central Tendency and Measure of Variability



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Measure of variability

- Mean deviation
- Variance
- Standard deviation
- Standard error of mean
- Range
- Less vs high value

- Pathogen is an infectious agent that feed on host body and cause damage to its host.

- Bacteria
- Viruses
- Protozoon

- **Phnetics methods** construct threes (phenogram) by considering the current states of characters without considering the evolutionary history that brought the species to their current phenotype.

Causes of speciation: 1=Geographical Isolation 2=Behavioural isolation 3=Reproductive isolation

[redacted]

What are the 7 classifications of a spider?

Classification of spiders



- Phylum - Arthropoda.
- Class - Arachnida.
- Order - Araenae.